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Method For Optimization Of Composition And Concentration Of Soil Solution Of Irrigated Soils For Nutrition Of Plants

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Abstract-The effect of the chemical composition and concentration of the soil solution of various agrofonds on the nutrient content in cotton of the Namangan-77 variety and its productivity, the changes in the chemical composition and concentration of the soil solution depending on the agricultural background and the season were studied.

As a result of studies, it was found that at a low concentration of soil solution the plant develops poorly, it cannot absorb a sufficient amount of nutrients and, conversely, a high concentration of the solution also negatively affects the plant. The optimal concentration is the most favorable environment for plant development, which helps to accelerate the absorption of nutrients by plants and ensure high crop yields.

The composition and reaction of the soil solution in the practice of agriculture is regulated by the application of fertilizers, soil cultivation, and land reclamation. For example, fertilizers are added to enrich the soil with nutrients, soil cleaning is carried out to eliminate salinization, etc.

The composition of the soil solution, the concentration and ratio of various compounds in it are subject to seasonal variation during the growing season. This is facilitated by the process of plant nutrition. Especially, in the middle of the growing season (July, August), noticeable changes occur in the composition of the soil solution.

The amount of nitrogen, phosphorus and potassium in the organs of cotton depends on environmental conditions. This directly affects the growth, development and productivity of cotton. Therefore, when 300 kg of nitrogen, 210 kg of phosphorus and 150 kg of potassium per hectare are used under the conditions of a typical serozem, optimal conditions for the growth and development of cotton variety Namangana-77 are created. This provides a high yield of cotton.

The results obtained are of great theoretical and practical importance in the knowledge of metabolic reactions in the soil-absorbing complex between the absorbed ions and ions in the solution; the action of fertilizers, irrigation and processing technologies on this process; scientifically sound application of fertilizers, as well as in the educational process of higher educational institutions.

Key words: soil, soil solution, anions, cations, concentration, agrotechnical measures, mineral fertilizers, agricultural background, soil-absorbing complex.

Introduction

Relevance of the topic. In agricultural production, the main means is soil. Soil properties, especially the optimization of soil solution for plant nutrition, are the main guarantee of obtaining a high and high-quality crop.

Soil solution plays an extremely important role in the process of soil formation, since all the processes of chemical and biological conversion of organic and mineral compounds are carried out with the direct participation of the liquid phase of the soil. This function of the soil solution can be generally called transformational. V.I.Vernadsky considered soil solutions to be one of the most important categories of natural waters, the main substrate of life, the main element of the biosphere mechanism. Importance of studies of pressurized soil solution natural and man-made factors is that the soil solution can be optimized for plant nutrition (Vernadsky et.al. 2019).

Despite the crucial role of soil solutions in the life of soils, their study is associated with numerous difficulties, due to the unresolved number of theoretical and methodological problems. Among researchers there is not even a single point of view on what, in fact, should be understood by the term "soil solution". There is no single standardized method for its preparation. The soil solution and its properties in the soils of Uzbekistan, especially in irrigated soils, have not been studied at all.

Currently, one of the main tasks of agriculture is to maintain the state of the soil solution for optimal nutrition of plants. In this regard, the study of the composition and concentration of soil solution, the creation of its favorable condition for plant nutrition is an urgent task of agriculture.

Francisco Javier Cabrera Corrala, Santiago Bonachela Castanoa, Maria Dolores Fernandez Fernandezb, Maria Rosa Granados Garciaa, Juan Carlos Lopez Hernandezb studied lysimetric methods for monitoring the conductivity of soil solution and nutrient concentration in greenhouse crops. He found a good correspondence between the concentration of nitrate, potassium, calcium and sodium solutions in the soil solution, measured using the rapid analysis system (Laqua ™), and measured using the reference laboratory method. This rapid system, in combination with the suction cup, can facilitate the farmers control of soil nutrient and salt status (Francisco et.al. 2016).

Kasper Reitzela, Benjamin L. Turnerb identifies pyrophosphate as the main constituent of soil phosphorus in lowland tropical rain forests and demonstrates that commercial pyrophosphatase can be used as a selective tool to quantify trace concentrations of pyrophosphate in soil solution (Kasper et.al. 2014).

According to Asgeir Rosseb Almas, Hilmar Thor Sævarsson, Tore Krogstada phosphorus (P) is an increasingly limited resource for food production, which requires the study of soil factors that control the rate of P release into the soil solution. Phosphorus use efficiency is among other, affected by the plants, the plant root induced displacement of P equilibrium in soil during uptake and the following rates of replenishment from geochemically active P stores in soil. Here, soil P partitioning and flux data collected from a 50 year field trial, based on application of differing P masses, was investigated using a diffusive gradients in thin films (DGT) technique and "DGT-induced fluxes in sediments" (DIFS) model. Partitioning of P in soil was accomplished by deploying DGT at increasing contact times (6 h to 120 h). DGT induced P fluxes increased with soil P, and they were highest the first hours after installation. A Langmuir adsorption approach to determine Kd concealed accumulated P labile stores, whereas Kd estimations from ammonium lactate (AL) extractions did not. The estimated fluxes in the latter situation stabilized after 24 h deployment, and they were more clearly reflecting the long term P treatments. The two Kd approaches showed the importance of considering inherent P-stores for calculating fluxes and deliveries of labile P available for plants during P-uptake. The use of DGT and DIFS in combination enabled good estimations of fluxes. Such fluxes may be used to estimate, In Situ, critical soil solution P concentrations available for plants during growth, in different soil types (Asgeir et.al. 2017).

Yao Yu Yanan Wan, Aboubacar Younoussa Camara, Huafen Li investigated the addition and aging corrections based on humic acids on the solubility of Cd in soil solution and its accumulation in rice. Yet, the effects of aging humic substances are not fully understood. A rice pot experiment was conducted to evaluate the effects of humic acid-based amendments on the mobility of Cd in soil solution and its uptake by rice when amendments were freshly added or aged for 130 d. The results showed that the aged and the unaged amendments generally decreased Cd concentration in soil solution, but the effect declined with time. Unaged HA-K (humic-potassium) reduced Cd concentration by 88% for the first sampling, but this dropped to 46% for the last sampling, compared to that of the control. All amendments, whether aged or not, reduced the content of Cd in rice seedlings, as well as in mature plants. Aged and unaged woody peat reduced the Cd content in seedlings by 79% and in grains by 70%, respectively. Aging of amendments caused lower pH and higher Cd concentration in the soil solution for all amendments and accordingly, the Cd content in rice seedlings or each part of mature plants in the aged group was higher than that of the unaged group. The applied amendments might reduce the solubility of Cd through the alteration in soil pH, and thus inhibit the uptake of Cd by rice, but the effects diminished with aging (Yao et.al. 2018).

Yusuke Unno, Hirofumi Tsukada, Akira Takeda, Yuichi Takaku, Shunichi Hisamatsu investigated the vertical distribution of the distribution coefficients of soil-soil solution (Kd) 125I, 137Cs and 85Sr in surface soils with a high content of organic substances and underground soils with a low content of organic substances on pastures and urban forests near spent nuclear fuel. Rokkasho Processing Plant, Japan. Kd of 137 Cs was highly correlated with water-extractable K^+. Kd of 85 Sr was highly correlated with water-extractable Ca^{2+} and SOC. Kd of 125 I− was low in organic-rich surface soil, high slightly below the surface, and lowest in the deepest soil. This kinked distribution pattern differed from the gradual decrease of the other radionuclides. The thickness of the high-125 I−Kd middle layer (i.e., with high radioiodide retention ability) differed between sites. Kd of 125 I− was significantly correlated with Kd of soil organic carbon. Our results also showed that the layer thickness is controlled by the ratio of Kd-OC between surface and subsurface soils. This finding suggests that the addition of SOC might prevent further radioiodide migration down the soil profile. As far as we know, this is the first report to show a strong correlation of a soil characteristic with Kd of 125I−. Further study is needed to clarify how radioiodide is retained and migrates in soil (Yusuke et.al. 2017).

Throughout Europe and the USA, forest ecosystem functioning has been impacted by long-term excessive deposition of acidifying compounds. Studies by Arne Verstraeten, Johan Neirynck, Gerrit Genouw, Nathalie Cools, Peter Roskams, Maarten Hens reported trends in the deposition of plantings and soil fluxes of inorganic compounds of nitrogen (N) and sulfur (S) over a 17-year period (1994-2010) at five ICP Forest monitoring sites in Flanders, northern Belgium. Deposition was dominated by N, and primarily NH_4^+. Deposition of SO_4^{2-} and NH_4^+ declined by 56-68% and 40–59% respectively. Deposition of NO_3^- decreased by 17-30% in deciduous forest plots, but remained stable in coniferous forest plots. The decrease of N and S deposition was parallelled by a simultaneous decline in base cation (BC = Ca^{2+} + K^{+} + Mg^{2+}) deposition, resulting in a 45-74% decrease of potentially acidifying deposition. Trends in soil solution fluxes of NH_4^+, NO_3^-, SO_4^{2-} and BC mirrored declining depositions. Nitrate losses below the rooting zone were eminent in both coniferous forest plots and in one deciduous forest plot, while net SO_4^{2-} release was observed in two deciduous forest plots. Critical limits for BC/Al ratio were exceeded at the three plots on sandy soils with lower cation exchange capacity and base saturation. Soil solution acid neutralizing capacity increased but remained negative, indicating that soil acidification continued, as the start of recovery was delayed by a simultaneous decrease of BC depositions and short-term soil buffering processes. Despite substantial reductions, current N deposition levels still exceed 4-8 times the critical load for safeguarding sensitive lichen species, and are still 22-69% above the critical load for maintaining ground vegetation diversity (Verstraeten et.al. 2012).

A study by José Bernardo Moraes Borin, Felipe de Campos Carmona, Ibanor Anghinoni, Amanda Posselt Martins, Isadora Rodrigues Jaeger, Elio Marcolin, Gustavo Cantori Hernandes, Estefânia Silva Camargo is aimed at assessing the electrochemical characteristics and the presence of the plant nutrient reaction in the soil, nutrient response irrigation rice water use in various flood management methods. For this, a field experiment was conducted in Southern Brazil, with three treatments: 1) continuous irrigation; 2) one water suppression (between V6–V8); and 3) two water suppressions (between V6–V8 and V8–V10). Regarding soil solution electrochemistry, the pH and redox potential (EH) were affected by the water suppression, increasing and decreasing due to soil reoxidation, respectively. The electrical conductivity (EC) decreased during the rice growing cycle, accompanying the plant development. The nutrients (except the potassium) were affected by water suppression, diminishing their availability. However, when the water layer was reestablished, there were no differences on soil solution electrochemistry among the irrigation methods. (Borin et.al. 2016).

Francisco Javier Cabrera Corral, Santiago Bonachela Castano, María Dolores Fernández Fernández, María Rosa Granados García, Juan Carlos López Hernández analyzes the conductivity dynamics (ECSS) and nutrient concentrations in a mixture of different amounts of soil and water in a solution of soil and water tension) in two greenhouse crops of tomatoes. ECSS obtained with zero tension lysimeters (funnel and plate lysimeter) was generally lower than that of a suction cup, regardless of soil depth. The ECSS obtained with zero-tension lysimeters (funnel and plate lysimeter) was generally lower than that with the suction cup, irrespective of soil depth. Moreover, the soil solution concentration of potassium, calcium, magnesium, sodium, chloride and sulphate obtained with funnel lysimeter (FullStop™) was generally lower than that with suction cup throughout both cycles, while no clear differences were found for the nitrate concentration at 0,25 m depth in the 2013/14 cycle or at 0,38 m depth in the 2015 one. Overall, it appears that the soil solutions collected with the suction cup and the funnel lysimeter represent different soil solution status and processes. The funnel lysimeter collects freely draining soil solution, and it may therefore provide better information about the movement of elements between soil horizons, whereas the suction cup can sample soil solution from soil pores with longer residence times, especially under unsaturated flow conditions, and might represent better the available element concentrations for plant nutrition studies. The differential response found for nitrate could be due to the fact that it is a very mobile element within the soil. The soil water matric potential was slightly higher in the soil with zero-tension lysimeters throughout most of the 2013/14 cycle, and so these devices might alter soil solution movement and water and nutrient availability. On the other hand, in general, a good fit was found between the soil solution concentration of nitrate, potassium, calcium and sodium measured with a rapid analysis system (Laqua™) and that measured using the reference laboratory method. This rapid system, in combination with the suction cup, can facilitate the farmers' control of soil nutrient and salt status (Francisco et.al. 2016).

Ricardo Perobelli Borba, Victor Sanches Ribeirinho, Otávio Antoniode Camargo, Cristiano Albertode Andrade, Carmen Silvia Kira, Aline Reneé Coscione monitored the soil solution (SS) for 10 years on a dark red oxysole clayey loamy clay / clay for agricultural purposes. SS was obtained by lysimeters installed along the walls of the well at a depth of 1 m, 2 m, 3 m, 4 m and 5 m. The major ions found in the SS were NO_3^-, SO_4^{2-}, Cl^-, Ca^{2+}, Mg^{2+}, Al^{3+}, Pb^{2+}, Cd^{2+} and Zn^{2+}, and the pH level ranged from 4 to 6.5 along the profile. Throughout the first three years of monitoring, the pH to a 3-m depth became more acidic, and in the last year, this trend reached 5 m. At the 5-m depth, the pH decreased from 6.5 to 4.5 from the first to the last monitoring. The SS acidification was provoked by both nitrite oxidation and ion leaching. The leaching of H^+ or the possible ion exchange/desorption of H^+ due to the leached cations (Ca^{2+} and Mg^{2+}) at the 4-m and 5-m depth caused the pH decrease. The ionic strength (IS) of the solution controlled the ion leaching. The sludge application increased the IS to 3 m, increasing the density of the soil charges and its ability to absorb ions. After the sludge application was completed, there was a decrease in IS of the SS as well as a decrease in ion absorption and retention abilities, which promoted leaching to greater depths. During the entire monitoring process, NO_3^-, Cd and Pb remained above the potability limit (Borba et.al. 2018).

Knowledge of the speciation of heavy metals and the role of dissolved organic matter (DOM) in soil solution is a key to understand metal mobility and ecotoxicity. In this study, soil column-Donnan membrane technique (SC-DMT) was used to measure metal speciation of Cd, Cu, Ni, Pb, and Zn in eighteen soil solutions, covering a wide range of metal sources and concentrations. DOM in these soil solutions was also fractionated into humic acid (HA), fulvic acid (FA), hydrophilic acid (Hy), and hydrophobic neutral organic matter (HON) by a rapid batch technique using DAX-8 resin (Zong-Ling et.al. 2015).

Object and subject of research. The object of research is the experimental section of the Department of Soil Science of the National University of Uzbekistan named after Mirzo Ulugbek. Subjects of research - Namangan-77 cotton variety, old-irrigated typical serozem, agrotechnical measures, mineral fertilizers, soil solution.

Purpose, research objectives and major versions. The main goal of the research is to develop a method for optimizing the soil solution of irrigated soils for plant nutrition based on the study of its composition and concentration. To achieve this goal, the following tasks were accomplished: conducting a field experiment on old irrigated typical serozem, determining the composition and concentration of soil solution, studying the influence of the chemical composition and concentration of the soil solution of various agrofonds on the nutrient content in cotton Namangan-77 and its yield, identifying changes in the chemical composition and concentration of soil solution depending on the agricultural background and the period of the year.

In natural soils, the soil solution is influenced by the solid, gas phases and natural vegetation cover. Whereas in irrigated soils, natural factors affecting the soil solution also include agricultural techniques (tillage, fertilizer use, irrigation, etc.).

When studying soil solution, one should not limit oneself to studying only the solid phase of the soil. It should be replenished with scientific materials about the liquid and gaseous phases of soils. Because all three soil phases are constantly interconnected. Especially from the point of view of agrochemistry and plant nutrition, all properties of the soil solution, such as concentration, chemical composition, osmotic pressure and their changes, should be known without exception. The vital activity of plants and microorganisms is also impossible without a soil solution, which performs for them both a protective and regulatory function, and is a source of nutrition.

The scientific novelty of the topic. For the first time under the conditions of an irrigated typical serozem, the composition and concentration of the soil solution were isolated and studied. The influence of the chemical composition and concentration of the soil solution of various agrofonds on the nutrient content in the cotton of the Namangan-77 variety and its productivity was determined, a change in the chemical composition and concentration of the soil solution depending on the agricultural background and season was revealed.

Research methodology

Studies on the isolation, determination of the composition, concentration of soil solutions and their dynamics are determined by the common methods in agrochemistry. To isolate and study the soil solutions, the Ischerekov-Komarova method of displacing the soil solution was used, where ethyl alcohol was used as the displacer.

Literature review

A number of foreign scientists (K.K.Gedroits, A.G.Doyarenko, A.A. Shmuk, S.A. Zakharov, A.A. Rode, made a significant contribution to the development of extraction methods, and especially in studying the composition and dynamics of soil solutions) P.A.Kryukov, N.A.Komarova, E.I.Shilova, A.A.Kizilova, N.J.Barrow, T.N.Choo). In the soils of Uzbekistan, especially in irrigated soils, the soil solution has not been studied at all, studies have not been conducted within the framework of certain programs, there is insufficient information on the composition of the soil solution.

Research results

Studies on the development of a method for optimizing the composition and concentration of the soil solution of the irrigated typical serozem for plant nutrition were carried out at the experimental site of the Department of Soil Science, located on the territory of the Botanical Educational and Scientific Center of the National University of Uzbekistan named after M.Ulugbek. Field experience consists of 4 options (table 1).

Table 1

Field experience diagram

				Phase 3-4		

№	Annual rate			Under the chil		With sowing			infusion shih leaf	Budding	Colorniya
	N	P	K	P	K	N	P	K	N	N	N
1	-	-	-	-	-	-	-	-	-	-	-
2	200	140	100	100	70	40	40	30	50	50	60
3	250	175	125	145	75	65	30	55	60	60	60
4	300	210	150	175	90	80	35	60	70	75	75

The soil of the experimental plot is characterized by a very low content in both the arable and sub-arable layer of humus, nitrogen, as well as the mobile forms of phosphorus and potassium (Table 2).

Table 2

Agrochemical properties of old irrigated typical serozem

Depth, cm	Humus, %	Total, %			Mobile, mg / kg		
		N	P	K	$N\text{-}NO_3$	P_2O_5	K_2O
0-30	1,13±0,02	0,118±0,002	0,165±0,003	1,51±0,02	27,7±0,3	35,5±0,2	215±3
30-50	0,85±0,01	0,085±0,002	0,143±0,002	1,06±0,03	20,5±0,2	18,8±0,3	140±4
50-69	0,80±0,02	0,080±0,001	0,120±0,003	1,01±0,01	11,5±0,4	13,1±0,1	130±2
69-99	0,67±0,02	0,050±0,002	0,109±0,004	0,77±0,02	7,9±0,2	5,5±0,3	70±3

The studied soils are carbonate. The upper boundary of the distribution of carbonates is 15-25 cm, and the lower is 70–120 cm (Fig. 1).

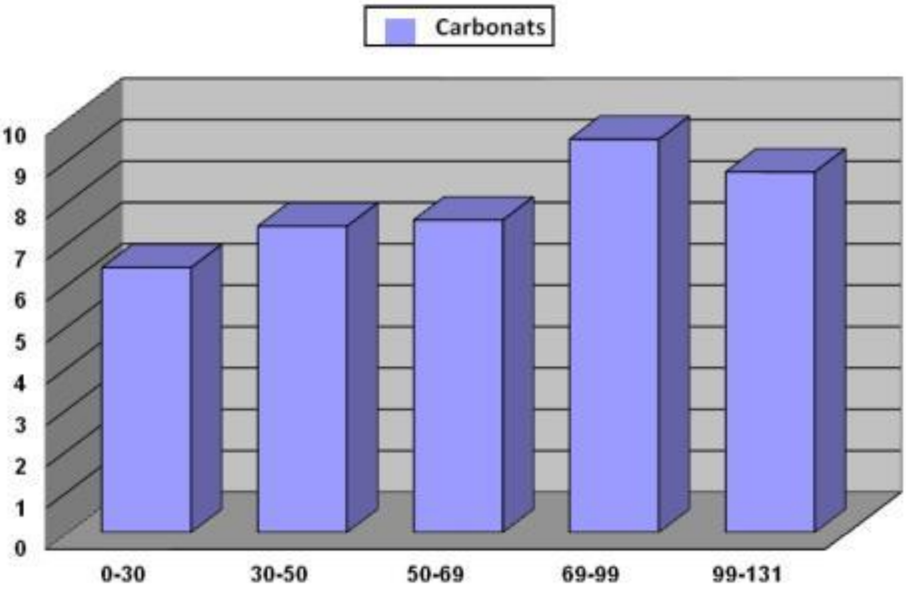

Fig. 1. **CO_2 carbonates in old irrigated typical sierozem**

We studied old-irrigated typical serozems in the soil solution, many compounds are in ionic form. Among them, Ca^{+2}, Mg^{+2}, and Na^{+} ions predominate. Ions K^{+}, NH_4^{+}, H^{+} are always present.

The data obtained show that the concentration of the soil solution of the old irrigated typical serozem ranges from 0,2-0,7 mg/l along the soil profile. Plants well absorb nutrients from the soil solution at a concentration of 0,2-1,0 mg/l. A further increase in concentration makes it difficult for nutrients to enter the plant.

In the studied old-irrigated typical serozem in the soil solution of anions, the content of normal carbonates along the soil profile ranges from 0,022-0,027%, ion chlorine 0,002-0,003%, ion sulfate 0,003-0,008%. Of the cations, sodium ions make up 0,002-0,005%, magnesium 0,001-0,003%, calcium 0,006-0,010% (table 3).

Table 3

Concentration of Old Irrigated Soil Solution typical serozem,%

Depth, cm	HCO_3^-	CL^-	SO_4^{-2}	Na^+	$Mд^{+2}$	Ca^{+2}
0-30	0,027	0,003	0,006	0,005	0,002	0,006
30-50	0,026	0,003	0,003	0,005	0,001	0,008
50-69	0,026	0,003	0,004	0,004	0,003	0,010
69-99	0,027	0,003	0,008	0,003	0,001	0,010
99-131	0,025	0,002	0,005	0,002	0,002	0,008
131-181	0,022	0,003	0,007	0,002	0,002	0,008

Table 4 shows the average chemical composition of the soil solution from the upper horizons of the old irrigated typical serozem, which were obtained from four options for the field experiment.

Table 4

The chemical composition of the soil solution in various agricultural backgrounds, mg/l

Options	HCO_3^-	Cl^-	NO_3^-	SO_4^{-2}	PO_4^{-3}	NH_4^+	Ca^{+2}	Mg^{+2}	K^+	Na^+
The control	48,2	2,2	9,3	4,0	0,27	5,0	39,1	26,1	2,1	29,1
N_{200} P_{140} K_{100}	51,3	2,5	10,2	4,1	0,30	6,0	43,4	30,7	2,2	32,0
N_{250} P_{175} K_{125}	57,5	2,6	11,4	4,2	0,31	7,5	45,8	34,8	2,3	34,3
N_{300} P_{210} K_{150}	60,7	2,7	12,0	4,1	0,40	7,9	49,5	37,4	2,4	35,8

From the data of table 4 it is seen that there is a noticeable difference between the experimental options in the composition of the soil solution. The composition of the soil solution of all options contains relatively more calcium, magnesium and potassium ions. The smallest number of anions of the cations was determined in the soil solution of the control variant. The use of fertilizers led to an increase in the content of anions of cations in the soil solution. In the arable soil horizon in the $N_{300}P_{210}K_{150}$ variant, the content of HCO_3 is 60,7 mg/l, NO_3-12,0 mg/l, NH_4-7,9 mg/l, Ca- 49,5 mg/l, Mg-37.4 mg/l, Na – 35,8 mg/l. Down the profile their number decreases. According to the results of field experiments conducted on typical irrigated soils, the concentration of soil solution changes during the growing season. The concentration was higher at the beginning of the growing season (Fig. 2).

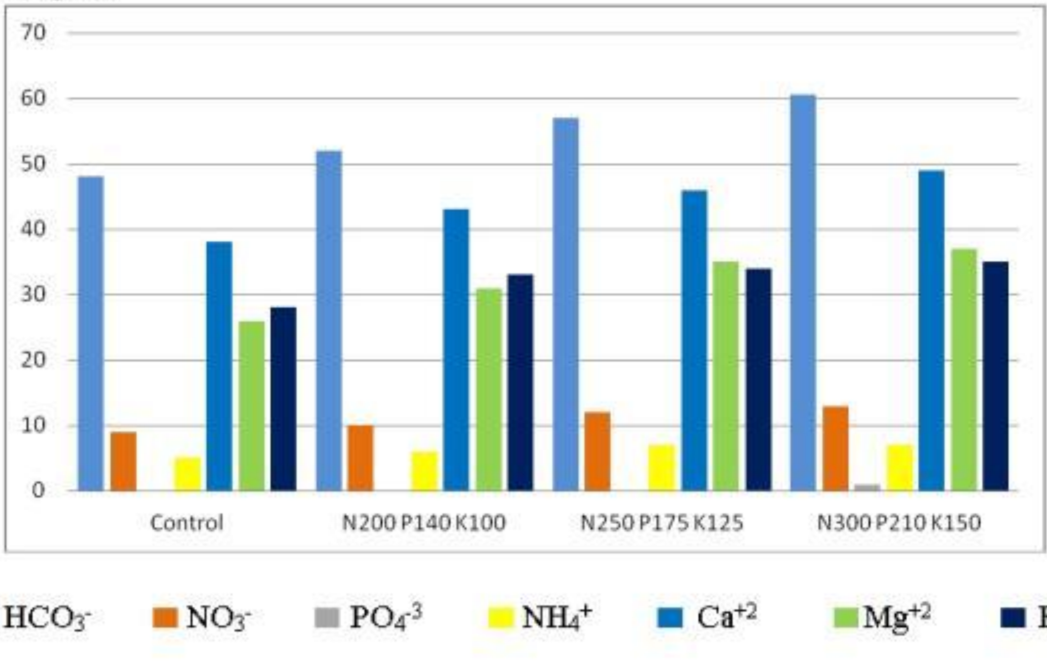

Fig. 2. **The concentration of soil solution (beginning of vegetation)**

The concentration of the solution decreases during plant development and intensive nutrition of elements (at the end of the growing season) by a small amount (Fig. 3).

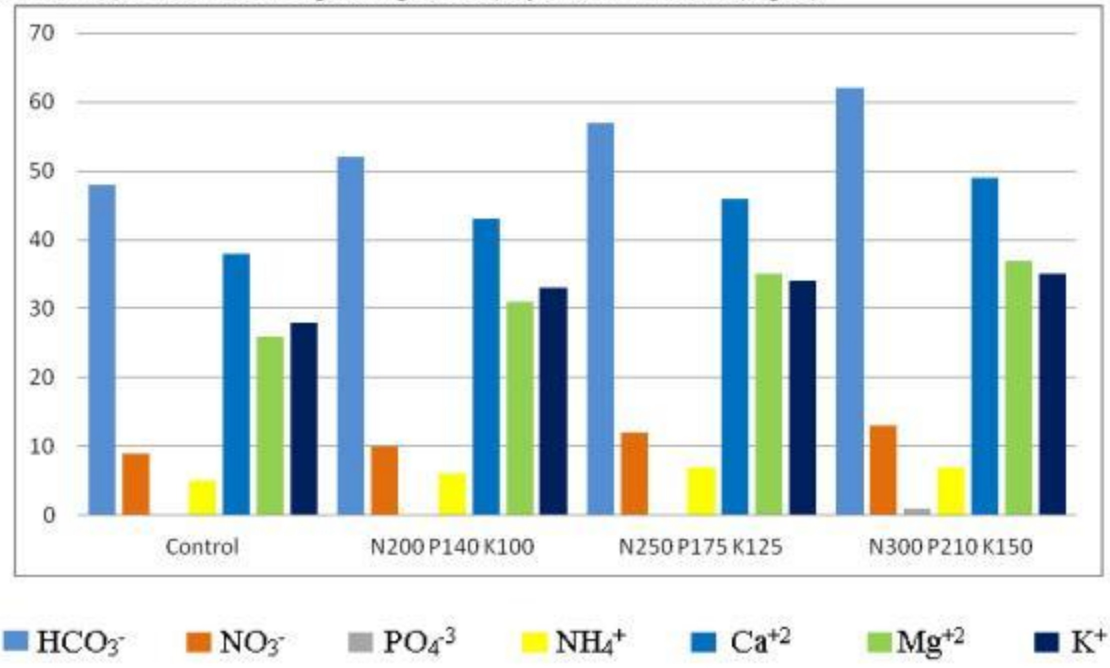

Fig. 3. **The concentration of soil solution (end of vegetation)**

The assimilation of nutrients by plants depends on its biology, the properties and composition of the soil, temperature, humidity, aeration, light, and, above all, the concentration of soil solution. The influence of the concentration and composition of the soil solution on various agricultural backgrounds on the amount of basic nutrients in the organs of cotton has been studied on old irrigated typical sierozems. For this purpose, plant samples were obtained during the budding period of cotton and chemical analyzes were performed.

Figures 4,5,6,7 show the amount of nutrients in the organs of cotton during budding. The results show that the low amount of nutrients in the organs of cotton is determined in the control variant of the experiment, where the concentration of soil solution is low. In fertilized versions, their amount increases, i.e. in the organs of cotton, the content of nitrogen, phosphorus and potassium increases. Therefore, since the use of fertilizers increases the amount of nutrients in the soil solution, cotton begins to absorb more nutrients through the roots and the size of the plant organ grows.

At the same time, plant nutrition can be differentiated by a quantitative indicator, that is, by the rate of assimilation of nutrients, as well as by quality, that is, by the ratio of assimilation of nutrients in different periods of the growing season.

The ratio of nutrients in the soil solution affects the amount of nitrogen in the cotton. For example, when the ratio of nitrogen to phosphorus is 1:0,7, the amount of nitrogen contained in the cotton is relatively small. When the amount of nitrogen and phosphorus was in the same proportion, it was noticed that the plant was more likely to absorb nitrogen. Thus, a high concentration of phosphorus in the soil solution contributes to a better absorption of nitrogen. This, in turn, activates the metabolism of the plant.

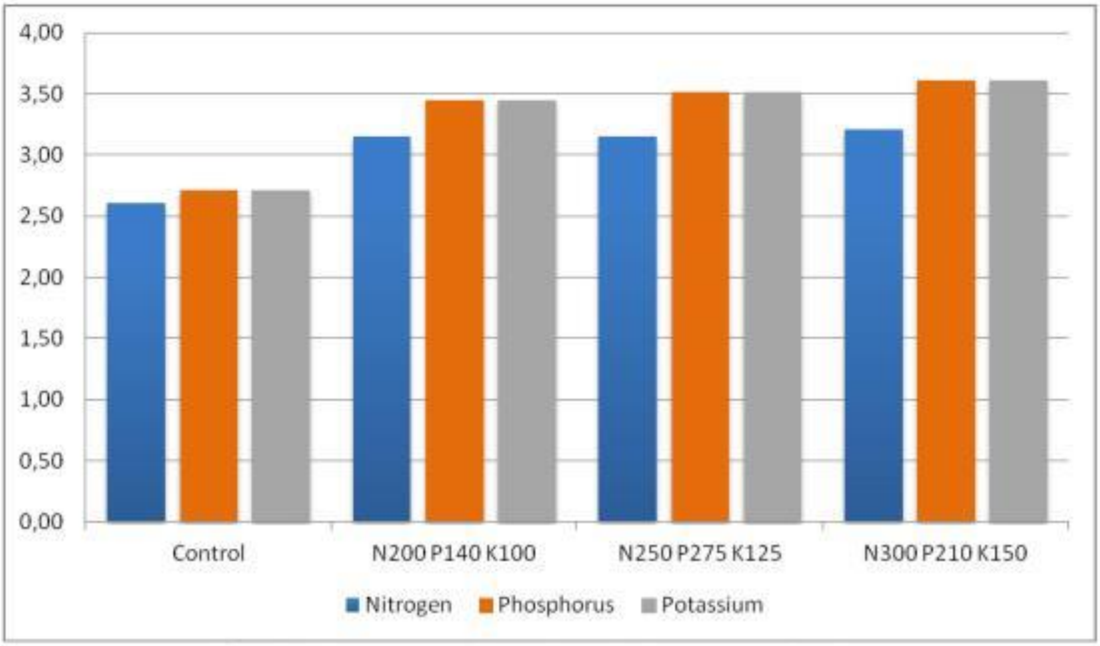

Fig. 4. **The content of nutrients in the leaves of cotton**

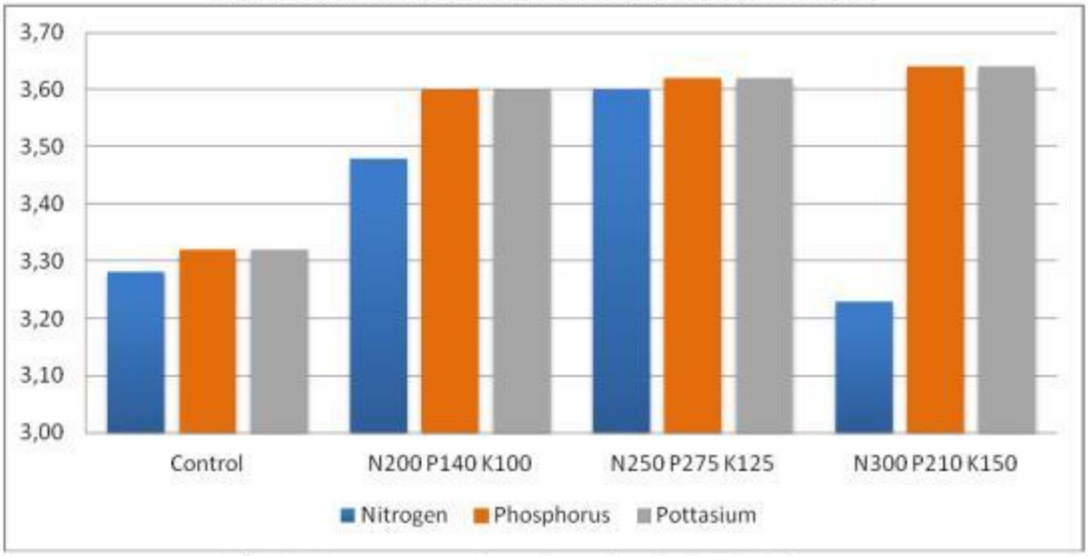

Fig. 5. **The content of nutrients in the buds of cotton**

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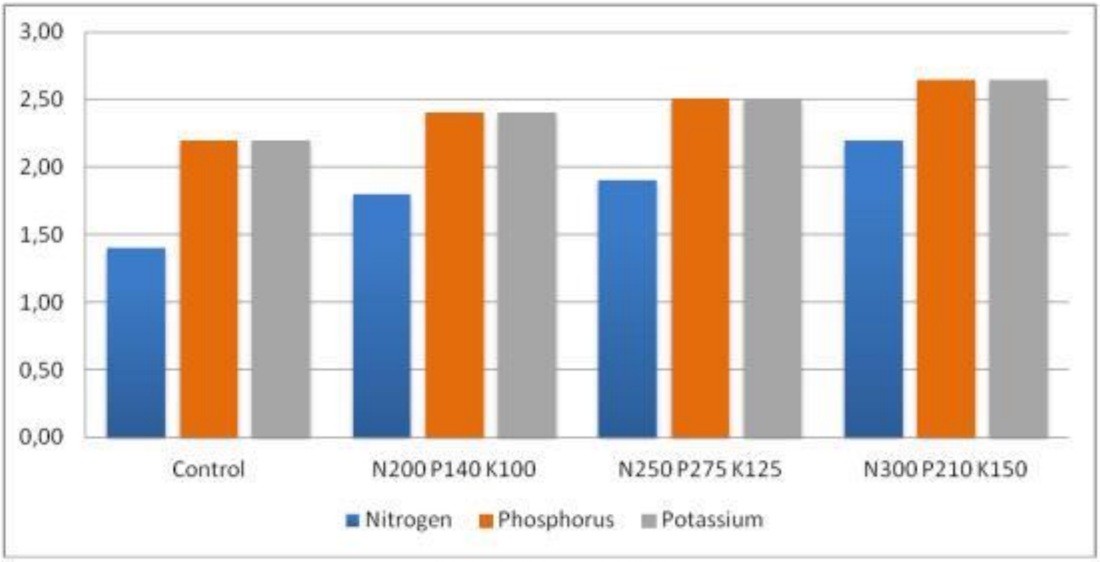
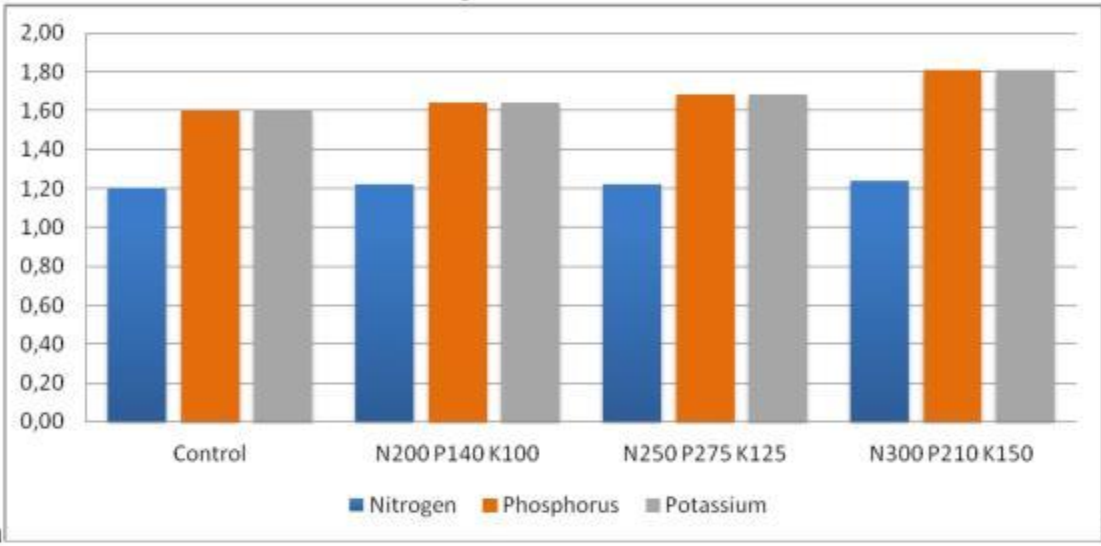


Fig.6. **The content of**



nu trie nts in the stalks of cotton

Fig. 7. **The content of nutrients in the roots of cotton**

The results obtained indicate that during the budding period, that is, in the active phase of physiological processes, the nutrient content in all organs of cotton is higher. The greatest amount of nutrients was observed in the cotton stalk, and then on the leaf. In the stem and roots, their amount is less than 40-50%.

Depending on the composition of the soil solution, it was found that the amount of nutrients in the organs of cotton, in particular the amount of phosphorus, was high in the buds and leaves of cotton. It was noted that the amount of phosphorus contained in cotton depends on the ratio of nutrients in the soil solution. In the variant where the ratio of nitrogen and phosphorus is 1:0,7, the amount of phosphorus in cotton is less, and in the variant with a ratio of 1:1 it is relatively larger. The greatest amount of potassium during the budding of the growing season, since nitrogen and phosphorus, was observed in the leaves and buds of cotton. Their number in the stem and roots of the plant is relatively small.

In the final phase of cotton growth, that is, at the end of the growing season, the amount of nutrients in the organs of the plant was closely varied depending on the concentration of the soil solution (Table 5).

Table 5

The effect of the concentration of soil solution of various agricultural backgrounds on the content of nutrients in cotton varieties Namangan-77, %

Options	Cotton Bodies	Nitrogen	Phosphorus	Potassium
The control	Leaves	1,57	0,40	1,28
	Stem	1,25	0,40	1,63
	Box	1,25	0,55	0,82
	Fiber	1,35	0,74	1,20
	Root	2,23	1,21	3,10
$N_{200} P_{140} K_{100}$	Leaves	1,80	0,99	2,48
	Stem	1,69	0,46	1,69
	Box	1,35	0,65	1,48
	Fiber	1,80	0,90	1,52
	Root	2,30	1,28	3,40
$N_{250} P_{175} K_{125}$	Leaves	1,92	1,02	2,52
	Stem	1,68	0,48	1,72
	Box	1,35	0,65	1,42
	Fiber	1,86	0,89	1,54
	Root	2,64	1,30	3,51
$N_{300} P_{210} K_{150}$	Leaves	2,00	1,03	2,75
	Stem	1,79	0,52	1,78
	Box	1,38	0,68	1,78
	Fiber	2,03	0,98	1,60
	Root	3,02	1,45	3,59

As can be seen from the data in table 5, by the end of the growing season, it was found that the amount of nitrogen, phosphorus and potassium in the organs of cotton decreased sharply. Especially the amount of nitrogen in the leaves is noticeably reduced. This can be explained by a decrease in the concentration of soil solution and reuse of nitrogen. By this time, the amount of nitrogen in the stem is reduced. However, it should be noted that the amount of nitrogen in the seeds is much higher.

The data obtained at the end of the growing season show that the phosphorus content in all organs of cotton has decreased as well as other elements. During this time, it was found that the amount of phosphorus is higher in the leaves and seeds of cotton, and less in the stem and root.

The total potassium content at the end of the growing season decreased significantly compared to other elements. However, the difference between the experimental options remains.

As you know, the metabolic process in plant organs slows down as the end of the growing season approaches. The nutrients in the leaf are used to synthesize organic matter. As a result, the amount of nutrients in the plant decreases. Depending on the use of nutrients from the soil solution, the removal of nitrogen, phosphorus and potassium also increases.

When studying the influence of norms and ratios of fertilizers on soil fertility and crop productivity, the main attention is paid to the ripening speed. This indicator is determined by harvesting over a period of time and comparing the results.

Table 6 shows the results of the harvest. According to the data obtained, unfertilized and mini-fertilized versions of the experiment, the buds ripen relatively faster. With an increase in fertilizer norms, the concentration of soil solution increases, which increases the yield of the crop, however, ripening is late. For example, in an unfertilized version, 50% of the cotton crop was harvested at the first harvest. On fertilized options, cotton harvested in the first crop is 10% less.

The lowest cotton yield was obtained in the variant with a low concentration of soil solution. The yield in this embodiment is 36,33 c/ga when using fertilizer in the norm of N_{250} P_{175} K_{125} per hectare. When fertilizer rates increased to N_{300} P_{210} K_{150}, the concentration of the solution increased and the yield increased to 40,21 c/ga, that is 22,41 c/ga higher than control.

Table 6

Effect of fertilizer rates and soil concentration cotton yield

Options	1-fee		2-fee		3-fee		General	
	c/ga	%	c/ga	%	c/ga	%	c/ga	%
The control	10,91	45,80	5,94	27,60	2,06	20,60	18,80	100
N_{200} P_{140} K_{100}	12,14	35,76	11,73	35,50	9,16	27,74	33,05	100
N_{250} $P_{175}K_{125}$	13,96	38,42	12,63	34,77	9,74	26,80	36,33	100
N_{300} P_{210} K_{150}	15,18	37,76	14,70	36,55	10,33	25,69	40,21	100

Thus, in fertilized versions, the opened boxes occur later. The main part of the total crop (70-75%) was harvested in the first and second harvests, and the smallest part (25-30%) was harvested in the third harvest.

In general, field experimental results show that at a solution concentration of 15,7-25,9 l/mmol, the conditions for plants were improved by improving plant development and the maximum yield was obtained at a concentration of 25,9 l/mmol. An increase in the concentration of the soil solution has a negative effect on the plant, causing the leaves to dry. Therefore, when forming a fertilizer system, this indicator must be taken into account, taking into account the different resistance of different cultures of the solution concentration.

Discussion

The soil of the experimental plot is characterized by a very low content in both the arable and the subsoil layer of humus, nitrogen, and also mobile forms of phosphorus and potassium.

We studied old-irrigated typical serozems in the soil solution, many compounds are in ionic form. Among them, Ca^{+2}, Mg^{+2}, and Na^{+} ions predominate. Ions K^{+}, NH_4^{+}, H^{+} are always present.

The data obtained show that the concentration of the soil solution of the old irrigated typical serozem ranges from 0,2-0,7 mg/l along the soil profile. Plants well absorb nutrients from the soil solution at a concentration of 0,2-1,0 mg/l. A further increase in concentration makes it difficult for nutrients to enter the plant.

As a result of studies, it was found that at a low concentration of soil solution the plant develops poorly, it cannot absorb a sufficient amount of nutrients and, conversely, a high concentration of the solution also negatively affects the plant. The optimal concentration is the most favorable environment for plant development, which helps to accelerate the absorption of nutrients by plants and ensure high crop yields. Changes in the composition and concentration of the soil solution have a direct effect on the assimilation of water and nutrients by plants.

The composition and reaction of the soil solution in the practice of agriculture is regulated by the application of fertilizers, soil cultivation, and land reclamation. For example, fertilizers are added to enrich the soil with nutrients, soil cleaning is carried out to eliminate salinization, etc.

The obtained data from the field experiment show that the composition of the soil solution, the concentration and ratio of various compounds in it are subject to seasonal variation during the growing season. This is facilitated by the process of plant nutrition. Especially, in the middle of the growing season (July, August), noticeable changes occur in the composition of the soil solution. During this growing season, the content of nutrients in the soil solution increases, and the drying function of the cotton root system is enhanced. This is due to the achievement of the nitrification process in the soil to the maximum point, an increase in the activity of phosphatase and the content of carbon dioxide in the soil air. As a result, the content of nitrogen, ammonium, and phosphorus in the soil solution increases. At the same time, significant changes in the soil environment will occur in the middle of summer. As a result, the environment changes to the alkaline or slightly acidic side.

In later periods of vegetation as a result of direct exposure to the plant, the amount of calcium ion in the soil solution decreases and the amount of potassium ions increases. As a result, the ratio of potassium ions to calcium ions expands. The process of entering nutrients into the root system depends on this ratio: the greater the ratio in the solution, the stronger the absorption capacity of the root or vice versa.

The data obtained show that the amount of nutrients contained in the organs of cotton is directly related to the composition and concentration of the soil solution that was created using fertilizers. At low concentrations of soil solution, a relatively small amount is observed, and vice versa, with a high concentration of soil solution, the amount of nutrients in the organs of cotton increases.

Thus, the amount of nitrogen, phosphorus and potassium in the organs of cotton depends on environmental conditions, the concentration of nutrients in the soil solution, the nutritional characteristics of the plant and its water supply. This directly affects the growth, development and productivity of cotton. Therefore, when 300 kg of nitrogen, 210 kg of phosphorus and 150 kg of potassium per hectare are used under the conditions of a typical serozem, optimal conditions for the growth and development of cotton variety Namangana-77 are created. This provides a high yield of cotton.

The results obtained are of great theoretical and practical importance in the knowledge of metabolic reactions in the soil-absorbing complex between the absorbed ions and ions in the solution; the action of fertilizers, irrigation and processing technologies on this process; scientifically sound application of fertilizers, as well as in the educational process of higher educational institutions.

Findings

Soil properties, especially the optimization of soil solution for plant nutrition, are the main guarantee of obtaining a high and high-quality crop.

Studies on the development of a method for optimizing the composition and concentration of the soil solution of the irrigated typical serozem for plant nutrition were carried out at the experimental site of the Department of Soil Science, located on the territory of the Botanical Educational and Scientific Center of the National University of Uzbekistan named after M.Ulugbek.

The soil of the experimental plot is characterized by a very low content in both the arable and the subsoil layer of humus, nitrogen, and also mobile forms of phosphorus and potassium. The studied soils are carbonate.

In the composition of the soil solution of the old irrigated typical serozem, many compounds are in ionic form. Among them, Ca^{+2}, Mg^{+2}, and Na^{+} ions predominate. Ions K^{+}, NH_4^{+}, H^{+} are always present. The concentration of the soil solution of the old irrigated typical serozem ranges from 0,2-0,7 mg/l along the soil profile.

In the soil solution of anions, the content of normal carbonates along the soil profile ranges from 0,022-0,027%, ion chlorine 0,002-0,003%, ion sulfate 0,003-0,008%. Of the cations, sodium ions make up 0,002-0,005%, magnesium 0,001-0,003%, calcium 0,006-0,010%.

The composition of the soil solution of all variants of the field experiment contains relatively more calcium, magnesium, and potassium ions. The smallest number of anions of the cations was determined in the soil solution of the control variant. The use of fertilizers led to an increase in the content of anions of cations in the soil solution. In the arable soil horizon in the $N_{300}P_{210}K_{150}$ variant, the content of HCO_3 is 60,7 mg/l, NO_3-12,0 mg/l, NH_4-7,9 mg/l, Ca 49,5 mg/l, Mg-37,4 mg/l, Na – 35,8 mg/l. Down the profile their number decreases.

According to the results of field experiments conducted on typical irrigated soils, the concentration of soil solution changes during the growing season. The concentration was higher at the beginning of the growing season.

Depending on the composition of the soil solution, it was found that the amount of nutrients in the organs of cotton, in particular the amount of phosphorus, was high in the buds and leaves of cotton. It was noted that the amount of phosphorus contained in cotton depends on the ratio of nutrients in the soil solution. In the variant where the ratio of nitrogen and phosphorus is 1: 0,7, the amount of phosphorus in cotton is less, and in the variant with a ratio of 1:1 it is relatively larger. The greatest amount of potassium during the budding of the growing season, since nitrogen and phosphorus, was observed in the leaves and buds of cotton. Their number in the stem and roots of the plant is relatively small.

In the final phase of cotton growth, that is, at the end of the growing season, the amount of nutrients in the organs of the plant was closely varied depending on the concentration of soil solution

On unfertilized and mini-fertilized versions of the experiment, the buds ripen relatively faster. With an increase in fertilizer norms, the concentration of soil solution increases, which increases the yield of the crop, however, ripening is late. For example, in an unfertilized version, 50% of the cotton crop was harvested at the first harvest. On fertilized options, cotton harvested in the first crop is 10% less.

The lowest cotton yield was obtained in the variant with a low concentration of soil solution. The yield in this embodiment is 36,33 kg /ga when using fertilizer in the norm of N_{250} P_{175} K_{125} per hectare. When fertilizer rates increased to N_{300} P_{210} K_{150}, the concentration of the solution increased and the yield increased to 40,21 c/ga, that is 22,41 c/ga higher than control.

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